

Exp no :08

Date :

Use Steg-Expose to Detect Hidden Data in Images

Description

StegExpose is a specialized tool used for **steganography analysis**. It works by evaluating the **statistical properties** of an image to estimate the probability of hidden data being embedded within it. This process helps forensic investigators detect steganography techniques like Least Significant Bit (LSB) embedding.

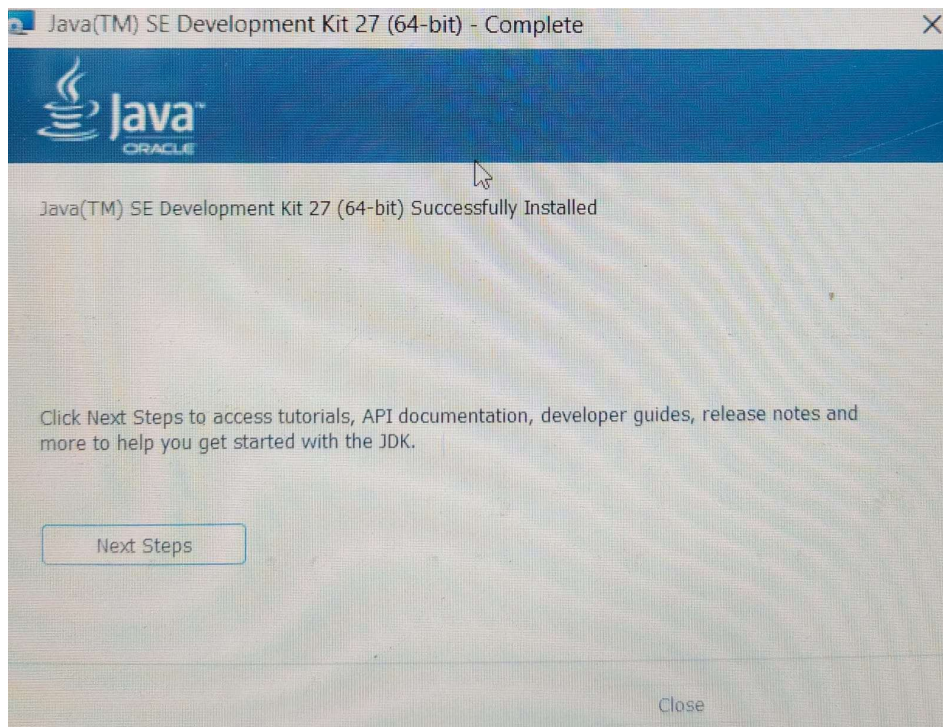
Prerequisites

- **Java Runtime Environment (JRE):** StegExpose is a Java-based application and requires JRE to run.
- **StegExpose Tool:** Download the latest .jar file from the official StegExpose GitHub repository.

Step-by-Step Process

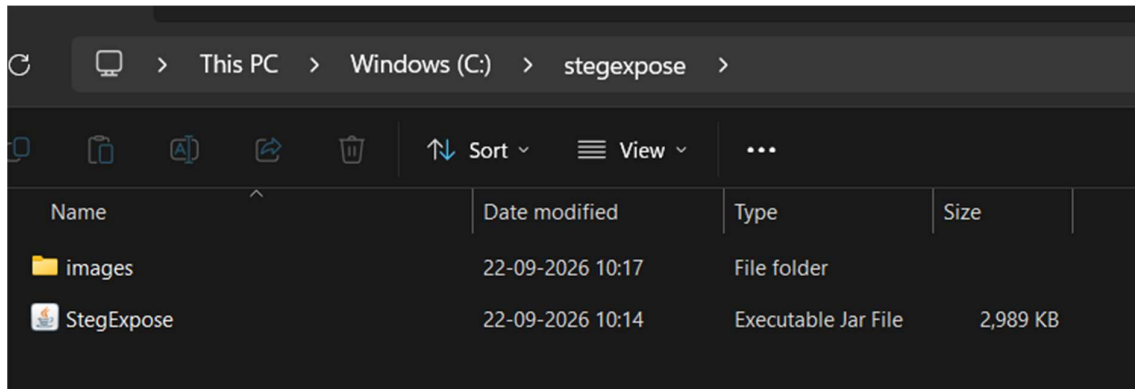
1. Download and Set Up StegExpose

1. Download the tool: Obtain the StegExpose.jar file from the GitHub repository.
2. Install Java: Ensure JRE is installed on your machine.
3. Prepare environment: Place the StegExpose.jar file in a dedicated working folder.



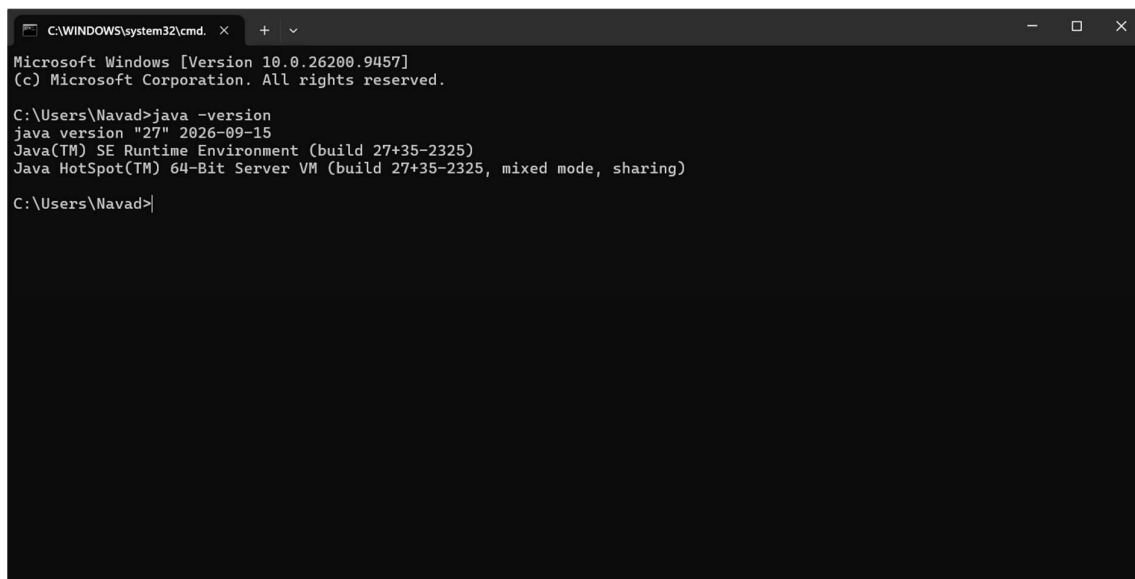
2. Select Images for Analysis

- Collect the images you suspect might contain hidden data.
- StegExpose supports common image formats such as .png, .jpg, and .bmp.



3. Open Command Line or Terminal

- Navigate to the folder where the StegExpose.jar file is located using your Command Prompt (Windows) or Terminal (Linux/macOS).



4. Run StegExpose on an Image

- Use the following command structure to analyze a single image for hidden data:

Command Structure:

`java -jar StegExpose.jar <image_file_path>` **Example:**

```
java -jar StegExpose.jar test_image.png
```

5. Analyze the Output

- StegExpose calculates a **"suspect" score** ranging from 0 to 1.
The higher the score, the more likely steganography is present.

Score Range	Interpretation (Suggested Thresholds)
Less than 0.2	Image is considered clean (no hidden data detected).
0.2 - 0.3	Possibly some hidden data is present.
Above 0.3	Likely that steganography is present.

- Example Output Analysis:** java -jar StegExpose.jar
suspect_image.png

Analyzing suspect_image.png...

Result: 0.4

Steganography likely present

```
C:\WINDOWS\system32\cmd. x + v
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Navad>C:\stegexpose\images\test.jpg
'C:\stegexpose\images\test.jpg' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\Navad>cd C:\stegexpose
C:\stegexpose>cd C:\stegexpose
C:\stegexpose>dir images
Volume in drive C is Windows
Volume Serial Number is CA9B-F22D

Directory of C:\stegexpose\images

22-09-2026  10:20    <DIR>          .
22-09-2026  10:17    <DIR>          ..
22-09-2026  10:18             20,196 Screenshot 2026-09-22 101810.png
               1 File(s)                20,196 bytes
               2 Dir(s)  23,568,683,008 bytes free

C:\stegexpose>|
```

```

C:\stegexpose>java -jar StegExpose.jar images default 0.2 results.csv

C:\stegexpose>dir
Volume in drive C is Windows
Volume Serial Number is CA9B-F22D

Directory of C:\stegexpose

22-09-2026  10:36    <DIR>          .
22-09-2026  10:20    <DIR>          images
22-09-2026  10:36                283 results.csv
22-09-2026  10:14           3,060,550 StegExpose.jar
                2 File(s)      3,060,833 bytes
                2 Dir(s)  23,586,791,424 bytes free

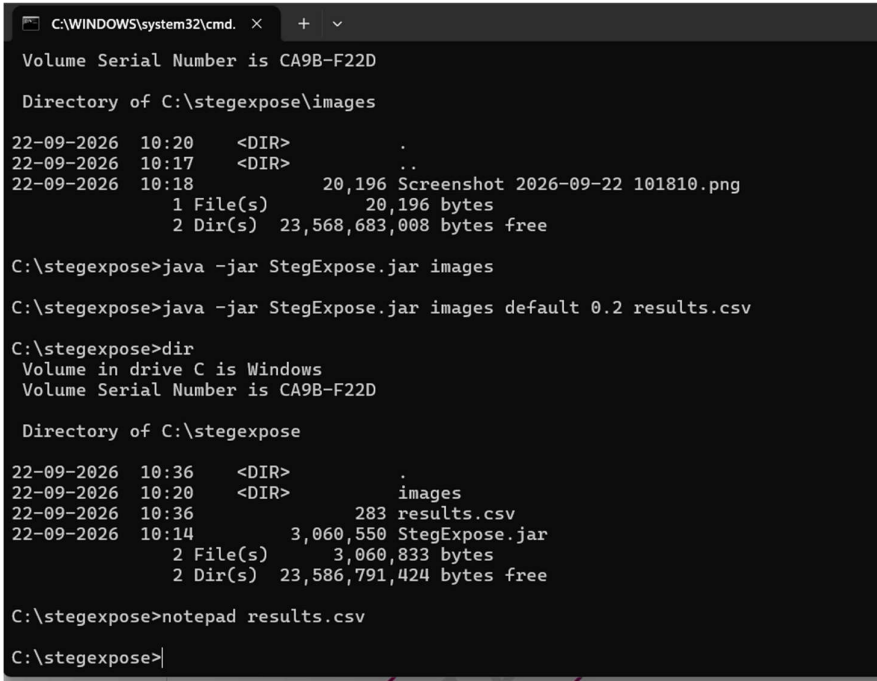
```

6. Batch Analysis (Multiple Images)

- To check multiple images at once, specify the folder path containing the images:

Command Structure:

```
java -jar StegExpose.jar <folder_path>
```



```

C:\WINDOWS\system32\cmd.  x  +  v
Volume Serial Number is CA9B-F22D

Directory of C:\stegexpose\images

22-09-2026  10:20    <DIR>          .
22-09-2026  10:17    <DIR>          ..
22-09-2026  10:18           20,196 Screenshot 2026-09-22 101810.png
                1 File(s)      20,196 bytes
                2 Dir(s)  23,568,683,008 bytes free

C:\stegexpose>java -jar StegExpose.jar images

C:\stegexpose>java -jar StegExpose.jar images default 0.2 results.csv

C:\stegexpose>dir
Volume in drive C is Windows
Volume Serial Number is CA9B-F22D

Directory of C:\stegexpose

22-09-2026  10:36    <DIR>          .
22-09-2026  10:20    <DIR>          images
22-09-2026  10:36                283 results.csv
22-09-2026  10:14           3,060,550 StegExpose.jar
                2 File(s)      3,060,833 bytes
                2 Dir(s)  23,586,791,424 bytes free

C:\stegexpose>notepad results.csv

C:\stegexpose>

```

7. Advanced Options (Optional)

- To view additional parameters, such as options for adjusting detection sensitivity or output verbosity, use the --help flag:

Command:

```
java -jar StegExpose.jar --help
```

8. Review the Results

- Review the scores generated for each image and use the threshold values to determine which images require further forensic investigation to extract the suspected hidden data.

Result:

The hidden data within the image was successfully detected using StegExpose. The analysis revealed that images with a suspect score above the threshold likely contain embedded steganographic content, confirming the tool's effectiveness in steganography detection.